REMARK ON OPENAI'S PROOF OF SAXL'S CONJECTURE: THE STAIRCASE SQUARE IN THE ALTERNATING GROUP

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Abstract. OpenAI (family 205 of `github.com/openai/math`) proved Saxl's conjecture and the tensor-square conjecture for symmetric groups. The motivating conjecture of Heide, Saxl, Tiep and Zalesski (HSTZ) concerns the alternating groups and remains open. We record three facts about the passage from S_n to A_n for self-conjugate labels. (i) We recall the closed formula for the square of a split character of A_n, which turns universality into the condition $g \neq |a|$ on the constituents with $a < 0$. By a known parity fact, the staircase split character is never universal when $m \equiv 2 \pmod 4$. (ii) Universality of χ^λ for S_n does not pass to its split constituents: for $\lambda = (4,4,4,3)$ the square of χ^λ contains every irreducible character of S_{15}, but the square of each constituent of $\chi^\lambda|_{A_{15}}$ misses the 14-dimensional standard character. (iii) By a direct modular computation of all Kronecker coefficients $g(\rho_m, \rho_m, \lambda)$, the HSTZ property holds for A_n, $n \in \{28, 36, 45, 66, 78\}$, with the staircase split character as witness. Previously published checks reached $n \leq 25$. We also explain why the multiplicity lower bounds contained in OpenAI's cyclic module, which suffice for small m, cannot be pushed to all m.

Attribution. Saxl's conjecture and the S_n tensor-square theorem are OpenAI's work (family 205, preprints *A Cyclic Polytabloid Proof of Saxl's Conjecture* and *Universal Tensor Squares for Symmetric Groups*, `github.com/openai/math`, commit `fd4aeeb2e`). Proposition 1 is classical in substance (Bessenrodt–Behns [1]; Pak–Panova–Vallejo [5, Main Lemma]); we include a proof because we need the exact multiplicities. Corollary 3 and the value $g(\lambda, \lambda, (n-1,1))$ used in Proposition 4 are in Bessenrodt–Bowman [3, Thm. 6.1, Cor. 6.2]. Developed with AI assistance (Anthropic Claude, Opus 5.5) under [AUTHOR NAME]'s direction; independent adversarial reviews were run by separate AI sessions and are not human peer review.

1. The square of a split character

Let $\lambda = \lambda' \vdash n$, let $\hat{h}(\lambda) = (h_1 > \cdots > h_d)$ be its diagonal hook lengths (all odd), $P = \prod h_i$ and $\varepsilon = \varepsilon_\lambda = (-1)^{(n-d)/2}$. Then $\chi^\lambda|_{A_n} = \chi_+ + \chi_-$, and $\chi_\pm$ differ only on the two A_n-classes $C_\pm$ of cycle type $\hat{h}(\lambda)$, where $\chi_\pm(C_\pm) = \frac{1}{2}(\varepsilon + \sqrt{\varepsilon P})$ and $\chi_\pm(C_\mp) = \frac{1}{2}(\varepsilon - \sqrt{\varepsilon P})$. In particular $\chi_\pm$ is real if and only if $\varepsilon = 1$. For $\nu \vdash n$ write $g_\nu = g(\lambda, \lambda, \nu)$ and $a_\nu = \chi^\nu(\hat{h}(\lambda))$.

Proposition 1. *The multiplicities of the irreducible characters of A_n in χ_+^2 are:*

(1) $\frac{1}{2}(g_\nu + \varepsilon a_\nu)$ *for* $\operatorname{Res}\chi^\nu$, *if* $\nu \neq \nu'$;
(2) $\frac{1}{4}(g_\nu + \varepsilon a_\nu)$ *for each constituent of* $\operatorname{Res}\chi^\nu$, *if* $\nu = \nu' \neq \lambda$;
(3) $\frac{1}{4}(g_\lambda + 1 + 2\varepsilon)$ *for* χ_+ *and* $\frac{1}{4}(g_\lambda + 1 - 2\varepsilon)$ *for* χ_-.

Proof. Write $\chi_\pm = A \pm \delta$ with $A = \frac{1}{2}\chi^\lambda|_{A_n}$ and δ supported on $C_+ \cup C_-$, $\delta(C_\pm) = \pm\frac{1}{2}\sqrt{\varepsilon P}$; thus $\delta^2 = \varepsilon P/4$ there. For a class function f of S_n, $\langle f|_{A_n}, 1\rangle_{A_n} = \langle f, 1\rangle_{S_n} + \langle f, \operatorname{sgn}\rangle_{S_n}$, and $|C_+ \cup C_-|/|A_n| = 2/P$. (1),(2): $\psi = \operatorname{Res}\chi^\nu$ is invariant under the outer automorphism, which swaps χ_+^2 and χ_-^2; hence $\langle \chi_+^2, \psi\rangle = \langle A^2 + \delta^2, \psi\rangle = \frac{1}{4}(g_\nu + g_{\nu'}) + \frac{2}{P} \cdot \frac{\varepsilon P}{4} a_\nu$, and $g_{\nu'} = g_\nu$ because $\lambda = \lambda'$. When $\nu = \nu' \neq \lambda$, χ_+^2 is constant on the two classes of type $\hat{h}(\nu) \neq \hat{h}(\lambda)$, so it pairs equally with the two constituents of ψ. (3): expand $(A+\delta)^2\overline{(A \pm \delta)}$; terms odd in δ cancel, $\langle A^3, 1\rangle = g_\lambda/4$, and $A\delta^2 = P/8$ on $C_+ \cup C_-$ (as $A = \varepsilon/2$ there). □

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Since multiplicities are non-negative integers, (1) recovers the bound $g_\nu \geq |a_\nu|$ and $g_\nu \equiv a_\nu \pmod 2$ of [5], and (3) gives $g_\lambda \equiv 1 \pmod 4$ (cf. [1]). Call χ_+ *universal* if χ_+^2 contains every irreducible character of A_n.

Corollary 2. *χ_+ is universal iff $\varepsilon = 1$, $g_\lambda \geq 5$, and $g_\nu > -a_\nu$ for every ν. If $g_\nu > 0$ for all ν (the S_n property) and $\varepsilon = 1$, this reduces to: $g_\nu \neq |a_\nu|$ whenever $a_\nu < 0$, and $g_\lambda \neq 1$.*

For the staircase ρ_m, $n = N_m = m(m+1)/2$, $\hat{h}(\rho_m) = (2m-1, 2m-5, \dots)$ has $d = \lceil m/2 \rceil$ parts and $(n-d)/2 = \sum_i (h_i - 1)/2 \equiv d(m-1) \pmod 2$.

Corollary 3 ([3, Cor. 6.2])**.** *$\varepsilon_{\rho_m} = -1$ iff $m \equiv 2 \pmod 4$. For such m the split staircase characters are non-real, so χ_+^2 does not contain the trivial character ($\langle \chi_+^2, 1\rangle = \langle \chi_+, \overline{\chi_+}\rangle = \langle \chi_+, \chi_-\rangle = 0$).*

So the staircase can be an HSTZ witness only if $m \not\equiv 2 \pmod 4$, i.e. $m = 3, 4, 5, 7, 8, 9, 11, 12, 13, \dots$

2. S_n-universality does not pass to A_n

Proposition 4. *Let $\lambda = ((q+s)^q, q^s)$ with $q, s \geq 1$; this is self-conjugate with exactly two part sizes. Then the split constituents $\chi_\pm$ of χ^λ are never universal: if $\varepsilon_\lambda = -1$ they miss the trivial character, and if $\varepsilon_\lambda = +1$ they miss* $\operatorname{Res} \chi^{(n-1,1)}$.

Proof. The case $\varepsilon = -1$ is Corollary 3's argument. Let $\nu = (n-1, 1)$. It is standard that $g(\lambda, \lambda, \nu)$ equals the number of distinct parts of λ minus one, here 1. The diagonal hooks are $h_i = 2(q+s-i)+1 \geq 2s+1 \geq 3$ $(1 \leq i \leq q)$, so the class $\hat{h}(\lambda)$ has no fixed points and $a_\nu = -1$. Proposition 1(1) gives multiplicity $(1-1)/2 = 0$. □

For $(q, s) = (1, 2)$ and $(3, 1)$, i.e. $\lambda = (3, 1, 1) \vdash 5$ and $\lambda = (4, 4, 4, 3) \vdash 15$, we have $\varepsilon = +1$, and a GAP computation shows $g(\lambda, \lambda, \nu) > 0$ for all ν. Hence a self-conjugate S_n-universal label need not give a universal character of A_n. The degenerate case $n = 5$ only says that the two 3-dimensional characters of A_5 are not universal. $(4, 4, 4, 3)$ is the first case with a non-hook label. Among all self-conjugate $\lambda \vdash n \leq 20$ with $\varepsilon = +1$, these are the only two that are S_n-universal but not A_n-universal (GAP, all $n \leq 20$; see the computation record). This does not refute HSTZ at $n = 15$: the staircase $(5, 4, 3, 2, 1)$ is a witness there.

3. HSTZ at triangular degrees $n \leq 78$

Theorem 5. *For $m \in \{3, 4, 5, 7, 8, 9, 11, 12\}$, i.e. $n \in \{6, 10, 15, 28, 36, 45, 66, 78\}$, the square of each irreducible constituent of $\chi^{\rho_m}|_{A_n}$ contains every irreducible character of A_n. In particular the HSTZ conjecture holds for these A_n.*

The cases $n \leq 25$ were already covered by computer checks (O'Brien for $n \leq 17$, reported in [4, Remark 1.3(3)]; Bessenrodt for $n \leq 25$, [2, §5]). To our knowledge (search of GitHub and arXiv up to 2026-10-09) the cases $n = 28, 36, 45, 66, 78$ are new.

Proof (computer-assisted). By Corollaries 2 and 3 ($\varepsilon = +1$), and because $g \geq |a|$ is automatic, it suffices to show the following for every $\lambda \vdash N_m$: $g_\lambda \neq -a_\lambda$ whenever $a_\lambda \leq 0$, together with $g(\rho_m, \rho_m, \rho_m) \neq 1$. A congruence $g \not\equiv c \pmod p$ proves $g \neq c$. We therefore compute the vector $(g(\rho_m, \rho_m, \lambda) \bmod p)_{\lambda \vdash N_m}$ for $p = 2^{31} - 1$, together with the exact integers $a_\lambda = \chi^\lambda(\hat{h}(\rho_m))$, for all $p(N_m)$ partitions (up to $p(78) = 12{,}132{,}164$). Since ρ_m is a 2-core, χ^{ρ_m} vanishes off the classes with all parts odd, so

$$\sum_\lambda g(\rho_m, \rho_m, \lambda) s_\lambda = \sum_{\mu \text{ odd}} \chi^{\rho_m}(\mu)^2 z_\mu^{-1} p_\mu.$$

We evaluate the right side in the Schur basis using the Murnaghan–Nakayama rule on β-sets. Each odd μ is split as head (parts $> T$) and tail (parts $\leq T$). The head vectors p_{head} are built along a trie. $\chi^{\rho_m}(\mu) = \langle p_{\text{head}}, Y_{\text{tail}}\rangle$ with $Y_{\text{tail}}[\nu] = \langle p_{\text{tail}} s_\nu, s_{\rho_m}\rangle$. The tail sum is evaluated by Horner's rule. For every m in the theorem the check finds no λ with $g \equiv -a$, and $g(\rho_m, \rho_m, \rho_m) \not\equiv 1$. The run counts are in Table 1. □

m	n	$p(n)$	$\#\{a<0\}$	$\#\{a=0\}$	tightest exact margin $\min_{a<0}(g-\lvert a\rvert)$
3	6	11	2	6	2 at $(4,2)$
4	10	42	10	21	2 at $(9,1)$
5	15	176	22	131	14 at $(13,2)$
7	28	3718	286	3145	34 at $(26,2)$
8	36	17977	1654	14656	6 at $(35,1)$
9	45	89134	4659	79813	62 at $(43,2)$
11	66	2323520	92010	2139531	(mod p only)
12	78	12132164	626548	10879571	(mod p only)

TABLE 1. Direct computation. For $m \le 9$ four primes and the CRT give the exact g (they are below $\dim \chi^{\rho_m}$, which is less than the product of the primes).

Validation. (a) For $m \le 6$ the full vectors agree with g computed from GAP's S_n character tables. (b) For $m \le 9$ the exact identity $\sum_\lambda g \cdot \dim \chi^\lambda = (\dim \chi^{\rho_m})^2$ holds over $\mathbb{Z}$, and for $m = 10, 11, 12$ it holds modulo p, as does $\sum_\lambda g\, a_\lambda = \chi^{\rho_m}(\hat{h})^2 = 1$. (c) Different head/tail thresholds T give bit-identical vectors. (d) $g(\rho_8, \rho_8, (8, 1^{28})) = 84465$, matching an earlier independent computation. (e) Proposition 1 was checked against direct decompositions of χ^2_+ in GAP's A_n tables for every self-conjugate $\lambda \vdash n$, $5 \le n \le 20$ (no mismatch). The tightest constraint in Table 1 is $\nu = (n-1, 1)$ or $(n-2, 2)$, e.g. $g(\rho_m, \rho_m, (n-1, 1)) = m - 1$ against $|a| = 1$ for even m.

4. What OpenAI's cyclic module gives, and where it stops

OpenAI's proof yields more than positivity. Let $W_m \subseteq S^{\rho_m} \otimes S^{\rho_m}$ be the cyclic module of [6, §3]. The *proof* of its Proposition 3.2 (an isomorphic image of $W_m \otimes \mathrm{sgn}$ contains a module that surjects onto M^{ρ_m}), Proposition 4.2 (band quotients $W_m \twoheadrightarrow \mathrm{Ind}(W_{m-s} \boxtimes U_{m,s})$, $s = 1, 2$) and Proposition 5.3 (every $\eta \vdash 2m-1$ with at most four rows occurs in $U_{m,2}$) give a recursive lower bound $w_m(\lambda)$ for the multiplicity of S^λ in W_m itself (each term bounds that multiplicity, which is what makes the recursion valid), hence

$$g(\rho_m, \rho_m, \lambda) \;\ge\; w_m(\lambda) := \max\Big(1,\; K_{\lambda', \rho_m},\; \sum_{\lambda/\nu \text{ horiz. } m\text{-strip}} w_{m-1}(\nu),\; \sum_{\nu \vdash N_{m-2}} w_{m-2}(\nu) \sum_{\ell(\eta) \le 4} c^\lambda_{\nu\eta}\Big).$$

Combined with Corollary 2, the test $\max(w_m(\lambda), w_m(\lambda')) \ge |a_\lambda| + 1$ for all λ with $a_\lambda < 0$ gives a second proof of Theorem 5 for small m. This second proof is conditional on the 205 propositions. We verified it independently for $m \le 8$; we also checked $w_m \le g$ for every λ there. An earlier internal computation pushed this route to $m = 12$, using an inductive surrogate for w that we did not re-run. The direct computation above makes it unnecessary.

We see no way to carry this route to all m, for two reasons. Neither is a proof of impossibility. The two-sided test itself passes on all hooks for $m \le 16$. *(1) One-sidedness on hooks.* For $\lambda = (m, 1^{N_m - m})$ and every $m \ge 5$, $w_m(\lambda) = 1$. Indeed $K_{\lambda', \rho_m} = \binom{m-1}{m-1} = 1$; the only horizontal m-strip leaves $(1^{N_{m-1}})$, whose bound is 1; and a width-two cut must remove $2m - 1 > m - 1 + 4$ boxes, so it would need a column of length > 4. The true value is large, e.g. $g = 84465$ at $m = 8$. The conjugate hook carries the information ($w = 263$ at $m = 8$), so the two-sided test passes. But any induction that feeds w_{m-1}, w_{m-2} back into the band sums sees only the weak side (computed for $m \le 16$). On hooks $\max_k |\chi^{(N-k, 1^k)}(\hat{h})|$ is $1, 4, 30, 289, 2974$ for $m = 8, 16, 24, 32, 40$. *(2) No mechanism for the margin.* Nothing in the construction relates the multiplicity of S^λ in W_m to $|\chi^\lambda(\hat{h})|$. The latter grows exponentially in m on hooks, so a uniform argument would need a new lemma (for instance, that hooks and near-hooks occur in W_m with exponential multiplicity). We do not have one. The direct method costs about $p(N_m)$ times a small polynomial; $m = 13$ ($n = 91$) is beyond our memory budget.

What is not claimed. HSTZ for non-triangular n, or for $m \equiv 2 \pmod 4$ (where a witness other than the staircase would be needed); any statement for general m. Theorem 5 for $n = 66, 78$ rests on one modular computation per n (with internal checksums), not on exact integers.

References

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[6] OpenAI, *A Cyclic Polytabloid Proof of Saxl's Conjecture* (2026), `github.com/openai/math`, family 205, commit `fd4aeeb2e`.

[7] OpenAI, *Universal Tensor Squares for Symmetric Groups* (2026), ibid.